SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

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CO2 - AT2 - SCENARIO BASED QUESTIONS

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QUESTIONS

Q1
Scenario Based - Smart Waste Recycling Plant System
25 marks

Q2
Scenario Based - IT Company Project Performance Evaluation (2025–2026).
25 marks

Q3
Scenario Based - 3. Semiconductor Chip Manufacturing System
25 marks

Q4
Scenario Based - 4. Cloud Storage Subscription Management (2026)
25 marks

4/4 answered

Q1 **Scenario Based - Smart Waste Recycling Plant System** 25 marks

A recycling plant processes 5,000 kg of waste every day.
Waste Categories:
Plastic
Paper
Metal
Electronic Waste
Efficiency Levels:
≥90% Recycled → Excellent
75–89% → Good
50–74% → Average
Below 50% → Poor
(a) Develop a C program using arithmetic operators to calculate recycled waste percentage and remaining waste. Use decision-making statements to determine plant efficiency.
(b) Use looping constructs to process waste data for 30 days. Use continue to skip maintenance days and break if the processing machine fails.

Your Code

```
#include<stdio.h>
void main()
{
    int totalwaste=5000, recycledwaste, rer
```

```
    float percentage;
    scanf("%d",&recycledwaste);
    percentage = (recycleedwaste*100.0)/t
    remainingwaste = totalwaste - recycle
    printf("\n recycled waste = %d kg\n",
    printf("%d\n",remainingwaste);
    printf("%.2f\n",percentage);
    if(percentage>=90)
        printf("plant efficiency: excelle
    else if(percentage>=75)
        printf("plant efficiency: good\n")
    else if(percentage>=50)
        printf("plant efficiency: average
    else
        printf("plant efficiency: poor\n")
        return 0;
}
```

Input

```
4600
```

Output

Run

Save